

Syllabus for Electromagnetic Theory: PHYS330

Dr. Jordan Hanson - Whittier College Dept. of Physics and Astronomy

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Abstract

This course provides a rigorous exploration of classical electrodynamics, including new computational electromagnetism (CEM) content. The course begins with a review of vector calculus and its physical interpretation in electromagnetism. Next, the electrostatics of charge distributions using Coulomb's law, Gauss's law, and the electric potential will be presented. The course proceeds to boundary-value problems and solutions to Laplace's equation, including multipole expansions and the method of images. Building on the introductory content, the course proceeds to properties and behavior of electric fields in matter, covering polarization, bound charges, and the electric displacement field. From electrostatics, the course proceeds to magnetostatics, introducing the Biot-Savart law, Ampère's law, vector potential, and the role of magnetization in materials. New CEM content serves as an introduction to time-dependent electromagnetic fields. Throughout the course, emphasis is placed on analytical problem-solving through active learning strategies. Final projects allow students to creatively explore scientific and engineering applications of electrodynamics and CEM.

Pre-requisites: PHYS150, PHYS180, PHYS185, PHYS275, MATH141, MATH142, MATH241

Course credits, Liberal Arts Categorization: 3.0 Credits, None

Regular course hours: TR 11:00am-12:20pm, SLC 232.

Instructor contact information: Discord: 918particle, Email: jhanson2@whittier.edu, Office: SLC 212.

Office hours: Open-door policy, or make an appointment via Discord/email.

Attendance/Absence: Students needing to reschedule midterms must notify the professor a few days in advance.

Late work policy: Late work is generally not accepted, but is left to the discretion of the instructor.

Text: Introduction to Electrodynamics, 5th ed. by D. Griffiths - <https://doi.org/10.1017/9781009397735>.

Grading: In-class warm-up exercises are marked only for attendance, and worth 10 percent of the grade. Weekly homework sets are intended to be challenging, and student collaboration is expected. Two short, open-book midterm exams will be completed outside of class. There is no final exam for the course. A final project and class presentation will be completed near the end of the semester. See Tab. 1.

Grade Settings: $< 60\% = F$, $\geq 60\%, < 70\% = D$, $\geq 70\%, < 80\% = C$, $\geq 80\%, < 90\% = B$, $\geq 90\%, < 100\% = A$.
Pluses and minuses: 0-3% minus, 3%-6% straight, 6%-10% plus (e.g. 79% = C+, 91% = A-)

Final projects: Students may contribute one of two types of final project: A) a more lengthy or challenging problem to teach the class, or B) a computational electromagnetism (CEM) demonstration. The final presentation should be approximately 20 minutes. Students should begin coding and producing basic solutions to challenge problems early in the course to facilitate creation of the final presentation.

ADA Statement on Disability Services: Whittier College is committed to make learning experiences as accessible as possible. If you experience physical or academic barriers due to a disability, you are encouraged to contact Student Disability Services (SDS) to discuss options. To learn more about academic accommodations, email disabilityservices@whittier.edu, call (562) 907-4825, or go to SDS which is located on the ground floor of Wardman Library.

Academic Honesty Policy: <http://www.whittier.edu/academics/academichonesty>

Course Objectives:

- To master the application of physical units, taking limits, and symmetry in physics
- To master techniques in mathematical physics and CEM and apply them to theoretical physics and engineering
- To use active learning techniques like directed warm-up exercises, peer learning, simulations, and student-designed final projects to build a better understanding of electrodynamics

Assignment	Weight	Assigned	Due
Problem Sets	40%	Alternate Fridays	Two weeks after assigned
Midterm 1	15%	Friday, December 5th, 2025	Monday, December 8th, 2025
Midterm 2	15%	Friday, December 5th, 2025	Monday, December 8th, 2025
Final Project Proposal	10%	Friday, October 31st, 2025	Friday, November 7th, 2025
Final Project Presentation	20%	Friday, October 31st, 2025	Thursday, December 4th, 2025

Table 1: Assignments for Electromagnetic Theory (PHYS330).

Course Outline:

1. Vector Analysis, Chapter 1

- (a) Vector algebra
- (b) Differential calculus in three dimensions
- (c) Integral calculus in three dimensions
- (d) Cylindrical and spherical coordinate systems
- (e) The Dirac Delta function
- (f) Theory of vector fields

2. Electrostatics, Chapter 2

- (a) The electric field
- (b) Divergence and curl of electrostatic fields
- (c) Electric potential
- (d) Work and energy in electrostatics
- (e) Conductors

3. Potentials, Chapter 3

- (a) Laplace's equation
- (b) The method of images
- (c) Separation of variables
- (d) Multipole expansion

4. Electric Fields in Matter, Chapter 4

- (a) Polarization
- (b) The field of a polarized object
- (c) The electric displacement
- (d) Linear dielectrics

5. Magnetostatics, Chapter 5

- (a) The Lorentz force law
- (b) The Biot-Savart law
- (c) The divergence and curl of magnetic fields
- (d) Magnetic vector potential

6. Magnetic Fields in Matter, Chapter 6

- (a) Magnetization
- (b) The field of a magnetized object
- (c) The auxiliary field
- (d) Linear and non-linear media

7. Electrodynamics, Chapter 7

- (a) The electromotive force
- (b) Electromagnetic induction
- (c) Maxwell's equations
- (d) The field of a rotating magnet

8. Computational Electromagnetism, CEM

- (a) The finite-difference time-domain (FDTD) technique
- (b) Defining boundary conditions in two and three dimensions
- (c) Defining radiating sources

(d) Examples and applications

- i. Simple waveguides and coaxial cables
- ii. Fiber optic couplers
- iii. Radio-frequency (RF) phased arrays
- iv. CEM enhanced with machine learning techniques
- v. Introduction to the Naval Engineering Education Consortium (NEEC) research at Whittier College and NSWC Corona